

Architectural and Design Document - Hikkaduwa Multi-Purpose Co-operative Society Ltd.
(HMCS)

Integrated Banking & Financial Management System

1. High-Level Architecture Overview

The system utilizes a **Microservices Architecture** to ensure scalability, fault tolerance, and independent deployment of banking modules across all 8 branches.

- **Frontend Layer:** React.js Single Page Application (SPA) providing a responsive, dashboard-driven web interface.
- **API Gateway:** Acts as the single-entry point for the frontend, handling routing, rate limiting, and CORS.
- **Security Layer:** JWT-based authentication and authorization using Spring Security, enforcing Role-Based Access Control (RBAC).
- **Microservices Layer:** Independent Spring Boot applications handling specific business domains.
- **Database Layer:** PostgreSQL databases. Each microservice uses a dedicated logical schema to avoid tight coupling and ensure data encapsulation.

2. Role-Based Access Control (RBAC) & Privilege Breakdown

The system requires strict access control based on user roles to ensure security and proper workflow delegation.

User Role	Access Scope	Key Privileges
System Administrator	Global	Full Access: Create/manage users, configure system parameters, trigger backups.
General Manager / Sec.	Global (All Branches)	Read/Approve: View consolidated cross-branch reports, final loan signature.
Branch Manager	Branch-Level	Manage/Approve: View branch data, approve special withdrawals, recommend loans.
Bank Service Manager	Branch-Level	Approve: Issue directives on loan applications, oversee branch compliance.

Loan Committee	Branch-Level	Read/Approve: View assigned loan applications and cast votes.
Senior/Field Officer	Branch-Level	Create/Read/Update: Enter member data, input loan details, upload valuations.
Teller / Counter Staff	Branch-Level	Execute: Process cash deposits/withdrawals, print passbooks, view basic profiles.
Valuer	Branch-Level	Execute: Input gold weight/purity, generate assessed values, issue pawn tickets.

3. Frontend Design: Role-Based Dashboard Views (React)

The React SPA dynamically routes users to their respective dashboards upon login, summarizing specific KPIs, pending tasks, and quick actions based on their role.

A. System Administrator Dashboard

- **Summary Widgets:** Total active users, system uptime, daily backup status, active branches (8/8 online).
- **Configuration Grid:** Adjust global system parameters (e.g., FD interest rate, Pawning interest rate).

B. General Manager / Secretary Dashboard

- **Summary Widgets:** Consolidated daily net cash position, total loan portfolio value across all 8 branches.
- **Approval Queue:** Final signature queue for multi-stage loan approvals.

C. Branch Manager Dashboard

- **Summary Widgets:** Branch EOD cash position, teller till summaries, total branch loans disbursed today.
- **Actionable Alerts:** Fixed Deposits approaching maturity, Pawning tickets nearing expiry.
- **Approval Queue:** Pending special withdrawals and loan recommendations.

D. Senior / Field Officer Dashboard

- **Summary Widgets:** Pending field visits, reports submitted this week.
- **Task Queue:** Pending asset investigation tasks.

- **Quick Actions:** Initiate new Loan Application, Register New Member.

E. Teller / Counter Staff Dashboard

- **Summary Widgets:** Personal till balance for the current shift.
- **Primary Interface:** Large "Search Member" bar.
- **Quick Actions:** Cash Deposit, Cash Withdrawal, Print Passbook.

F. Valuer (Thaksarukaruge) Dashboard

- **Summary Widgets:** Total gold valued today, active pawn tickets, tickets overdue for auction.
- **Pawning Calculator:** Auto-calculates Assessed Value and Advance Amount based on weight and purity inputs.

4. Microservices Breakdown & API Specifications (Spring Boot)

The backend is divided into specialized domain-driven microservices.

4.1 Identity & Access Service (hmcs-auth-service)

- Handles login and issues context-aware JWTs.
- POST /api/v1/auth/login - Authenticate user and return JWT.

4.2 Member & Account Service (hmcs-account-service)

- Manages profiles, Savings, Children's accounts, and FDs. Powers the Teller and Field Officer dashboards.
- POST /api/v1/members - Register a new member.
- GET /api/v1/members/search?q={query} - Search member details.
- POST /api/v1/accounts - Open a new account.
- POST /api/v1/transactions/deposit - Process cash deposit.
- POST /api/v1/transactions/withdraw - Process cash withdrawal.

4.3 Loan Origination & Workflow Service (hmcs-loan-service)

- Handles the multi-stage approval workflow and EMI tracking.
- POST /api/v1/loans/apply - Submit a loan application.
- GET /api/v1/loans/queue?stage={stageId} - Fetch tasks for approval dashboards.

- PUT /api/v1/loans/{loanId}/stage/{stageId}/approve - Progress loan to next stage.
- GET /api/v1/loans/{loanId}/schedule - Generate EMI repayment schedule.

4.4 Pawning Service (hmcs-pawning-service)

- Manages the gold loan lifecycle.
- POST /api/v1/pawning/issue - Issue a digital pawn ticket.
- POST /api/v1/pawning/tickets/{ticketId}/redeem - Process redemption.
- GET /api/v1/pawning/overdue - Fetch pawn tickets exceeding the 1-year limit.

4.5 Reporting & Aggregation Service (hmcs-reporting-service)

- Generates EOD reports without impacting transaction performance.
- GET /api/v1/reports/eod/consolidated - For General Manager (all branches).
- GET /api/v1/reports/eod/branch/{branchId} - For Branch Manager.

5. PostgreSQL Database Setup & Schema

This script utilizes PostgreSQL schemas to logically isolate data for each microservice within a unified database.